

Kelly Ryoo

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EDUCATION

Cornell University, College of Arts and Sciences

May 2024

Bachelor of Arts in Computer Science, Literatures in English

GPA: 3.8/4.0 (Dean's List)

Coursework: Analysis of Algorithms, Intro to Artificial Intelligence, Intro to Machine Learning, Language and Information, Operating Systems, Computer System Organization and Programming, Functional Programming, Discrete Structures, Object Oriented Programming and Data Structures

EXPERIENCE

Cisco Systems

New York, NY

Software Engineer

September 2024 ~ Present

- Engineered an end-to-end agentic AI NL2SQL pipeline (LangGraph, LLM, Python) with custom prompt engineering to help 20K+ company sellers query business data without SQL knowledge, with automated retry and LLM-driven query correction on failure
- Built a RAG-backed context pipeline for the NL2SQL system using pgvector similarity search over a PostgreSQL knowledge base of schemas, business glossary, and example queries, with Redis for conversational memory
- Designed MCP servers and a Python REST wrapper API around Helios, an internal AI assistant, enabling engineering teams to build and extend custom chatbot applications on top of existing AI infrastructure

Cornell University

Ithaca, NY

Teaching Assistant

January to May 2024

- Supported 100+ students in CS 4300 (Language and Information); hosted weekly office hours, assessed homeworks and final projects, advised and mentored a team of students about information retrieval systems (Python) and web development

Cisco Systems

San Jose, CA

Software Engineering Intern

June 2023 ~ August 2023

- Analyzed lower-level forecast data and built out two machine learning models (Linear Regression, Neural Network) in Python and tensorflow to predict company-wide earnings and improve forecasting process
- Automated product test cases for NGF using Cucumber and Selenium, writing 26 scenarios, 11 step definitions, and 18 Java functions, thereby reducing manual testing effort significantly

McMahon Lab

Ithaca, NY

Undergraduate Researcher

January 2021 ~ October 2021

- Programmed lab laser instruments with Python to enable experiments and collection of data
- Formulated experiments with the RSA (Real Time Spectrum Analyzer) machine and wrote Python code to collect data during swept mode

University of Missouri

Columbia, MO

Research Intern

May 2021 ~ July 2021

- Conducted literature reviews of 10 research papers about bed sensors and sleep data machine learning algorithms
 - Collaborated with two other interns to design and program a deep learning (CNN-LSTM hybrid) model that classifies polysomnography data, achieving a noninvasive solution to monitor sleep quality and related disorders
 - Attained an 85% accuracy rate using PyTorch to preprocess, extract features, and classify sleep stages
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PROJECTS

Plant Pals

Spring 2023

- Designed and programmed an online information retrieval system that allows users to query plant information
- Implemented Jaccard similarity, Cosine similarity, and Rocchio's relevance feedback algorithms in Python
- Built the front-end layout and website features using HTML, CSS, and Javascript

Podcast Website ([link](#))

Summer 2021

- Designed the website layout, user interface, and visual graphics for the podcast "Learn Korean and Korean Culture" to serve over 90,000 listeners, then programmed the website using React, HTML, CSS, and Bootstrap
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SKILLS & INTERESTS

Programming: Python, Java, SQL (Oracle, PostgreSQL), JavaScript, React, HTML/CSS, C, OCaml, R

Tools & Skills: LangGraph, LangChain, FastAPI, REST APIs, Flask, vector search, Redis, Git, Selenium, Cucumber, Pytorch, Bootstrap, Anthropic, OpenAI, NLP, Agile, RAG, LLM

Interests: AI and technology policy, large language models, digital humanities, literature, social impacts of tech